Probability
Lecture 9: Central Limit Theorems

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Content

1. Central Limit Theorems
2. Applications

Law of Large Numbers

- Recall that: for sample data $\{X_1, \ldots, X_N\}$ **i.i.d** (independent identically distribution, the sample average is defined as

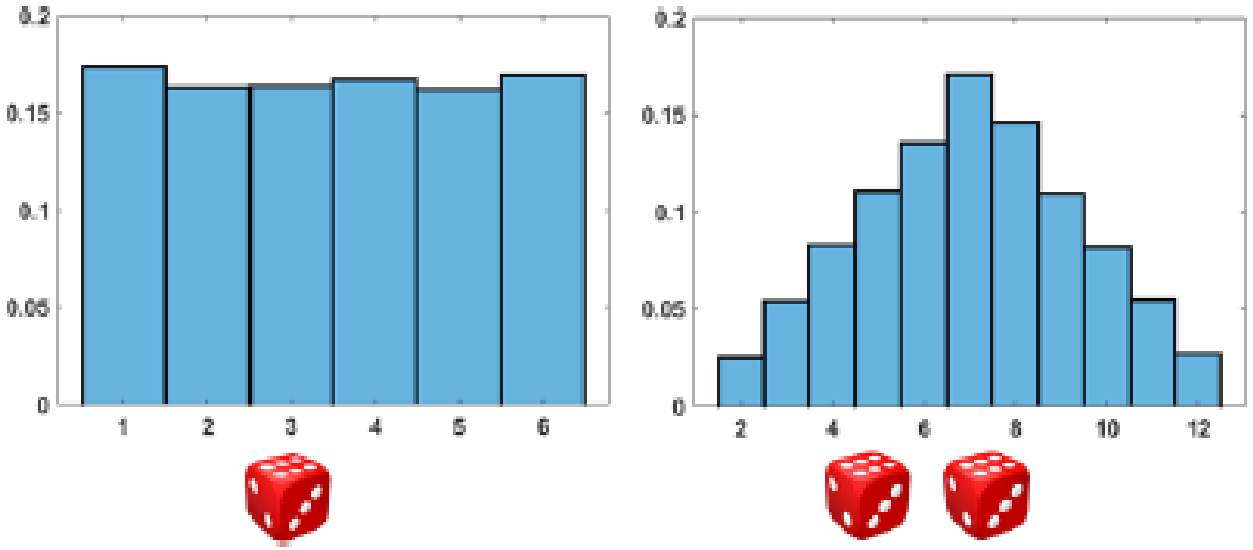
$$\overline{X} = \frac{X_1 + X_2 + \ldots + X_N}{N}.$$

- Law of Large Numbers

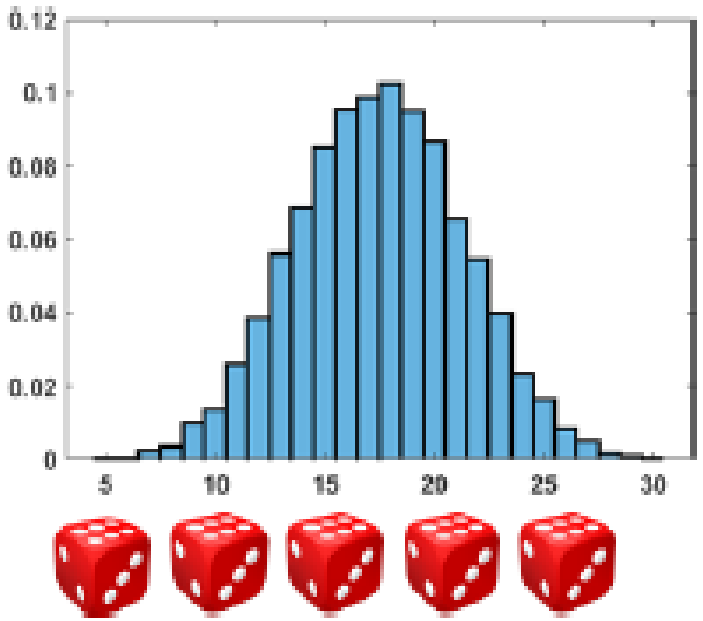
$$\overline{X} \overset{\mathbb{P}(a.s)}{\to} \mu = \mathbb{E}[X_1].$$

- Applications in statistics: average height, weight, ...
- Questions:
  - Is your estimation good (sharp)?
  - How many samples are there necessary?

Numerical simulation

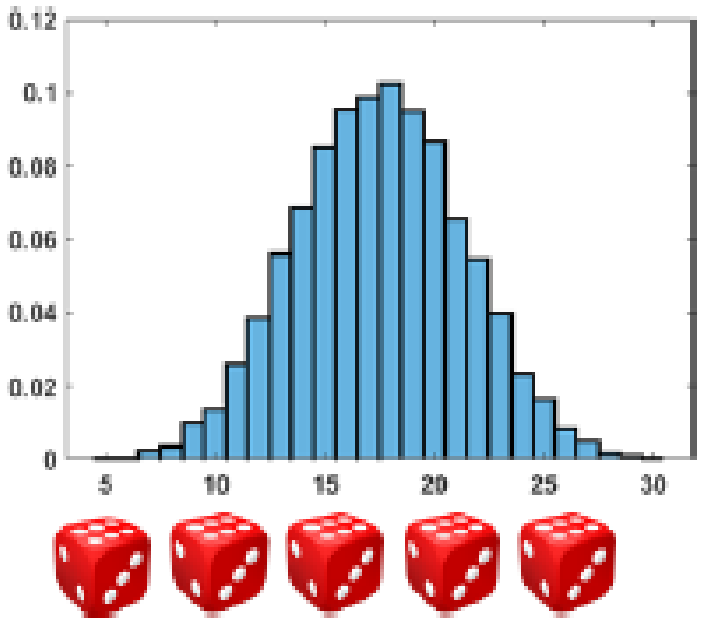


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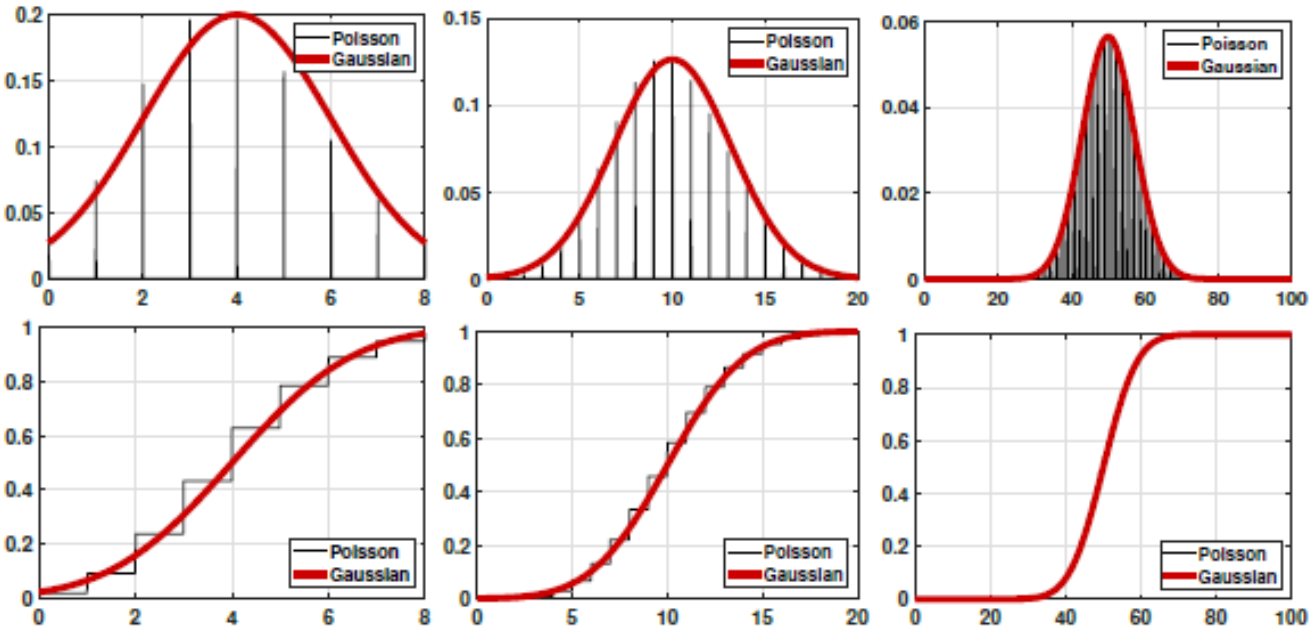
Observation: close to normal distribution!

Numerical simulation



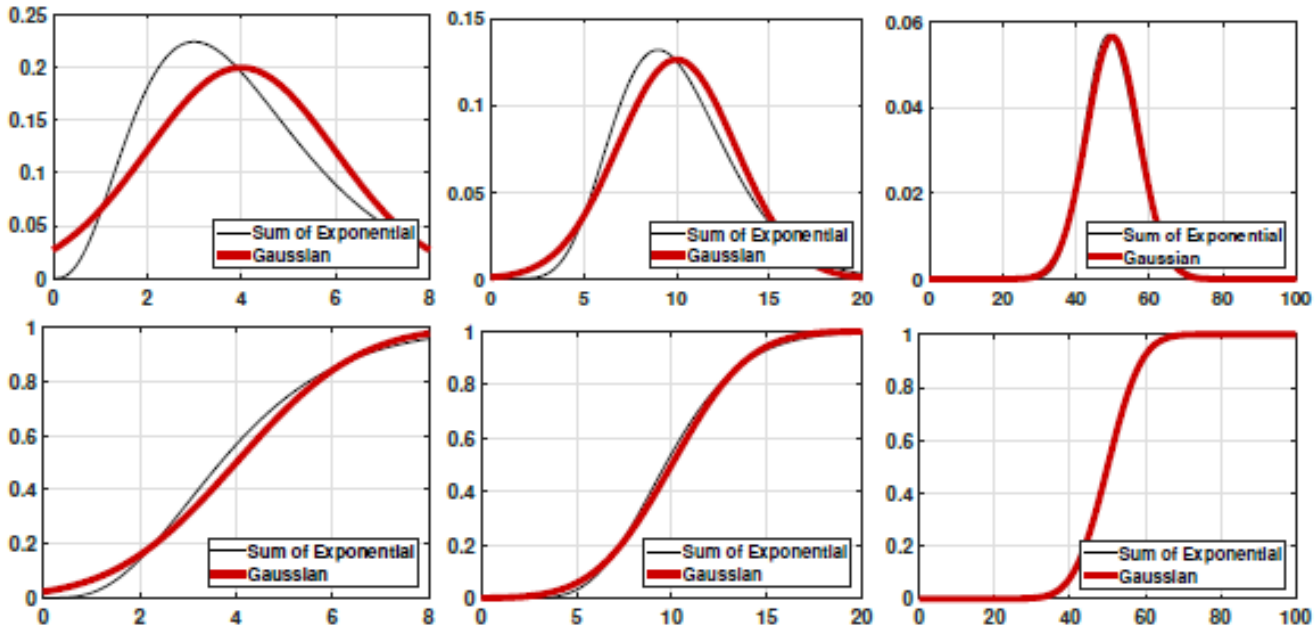
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Numerical simulation



(a) $N = 4$ (b) $N = 10$ (c) $N = 50$

Numerical simulation



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Convergence in distribution

Definition

Let $Z_1, \ldots, Z_n, \ldots$ be random variables with CDFs $F_{Z_1}, F_{Z_2}, \ldots$. We say that the sequence Z_n **converges in distribution (in law)** to a random variable Z with CDF F_Z if

$$\lim_{n \to \infty} F_{Z_n}(t) = F_Z(t),$$

for every continuous point t of F_Z. We denote as

$$Z_n \overset{d(\mathcal{L})}{\to} Z.$$

Central Limit Theorem

Theorem

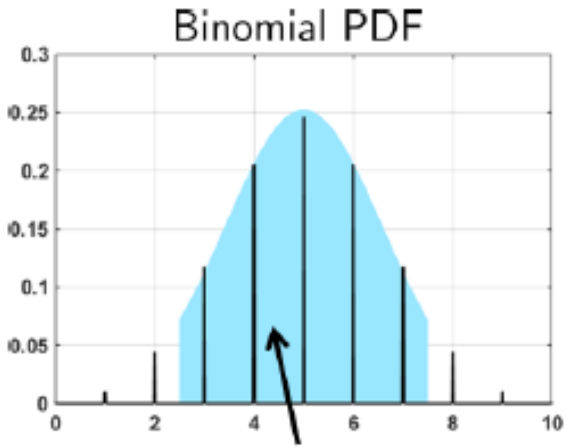
Let $X_1, X_2, \ldots$ be i.i.d. random variables of mean $\mathbb{E}[X_1] = \mu$ and variance $Var(X_n) = \sigma^2$. Then

$$Z_n = \sqrt{n}\frac{\overline{X} - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1) = Z,$$

i.e.

$$\lim_{n\to\infty} F_{Z_n}(t) = F_Z(t) = \int_{-\infty}^{t} \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx.$$

Idea



$\mathbb{P}[a \leq \overline{X}_N \leq b]$ = sum of deltas

CLT says

$\approx$



$\mathbb{P}[a \leq \overline{X}_N \leq b]$ = integrate the curve

Proof of CLT: moment-generating function

S. Chan assume more that $\mathbb{E}[|X_n|^3] < \infty$.

$$M_{Z_n}(s) = \mathbb{E}[e^{sZ_n}] = \mathbb{E}[\exp\left(s\sqrt{n}\frac{\overline{X}-\mu}{\sigma}\right)]$$
$$= \prod_{i=1}^{n} \mathbb{E}[\exp\left(s\frac{X_i-\mu}{\sqrt{n}\sigma}\right)]$$

by Taylor expansion,

$$= \prod_{i=1}^{n} \mathbb{E}\left[1 + \frac{s}{\sqrt{n}\sigma}(X_i-\mu) + \frac{s^2}{2\sigma^2 n}(X_i-\mu)^2 + O\left(\frac{|X_i-\mu|^3}{\sigma^3 n^{3/2}}\right)\right]$$
$$= \prod_{i=1}^{n} \left[1 + \mathbb{E}\frac{s}{\sqrt{n}\sigma}(X_i-\mu) + \mathbb{E}\frac{s^2}{2\sigma^2 n}(X_i-\mu)^2\right] = \left(1+\frac{s^2}{2n}\right)^n$$
$$\to e^{s^2/2} = M_Z(s).$$

Proof of CLT: Characteristic functions

$$\begin{aligned}\varphi_{Z_n}(t) =& \mathbb{E}\exp\left(it\sum_{j=1}^{n}\frac{X_j-\mu}{\sigma\sqrt{n}}\right)\\ =& \prod_{j=1}^{n}\mathbb{E}\exp\left(it\frac{X_j-\mu}{\sigma\sqrt{n}}\right) = \left(\varphi_X(\frac{t}{\sigma\sqrt{n}})\right)^n.\end{aligned}$$

We have

$$\varphi_X(t) = 1 - \frac{\sigma^2 t^2}{2} + t^2\alpha(t).$$

Hence

$$\log\varphi_{Z_n}(t) = n\log\left(1 - \frac{t^2}{2n} + \frac{t^2}{n\sigma^2}\alpha(\frac{t}{\sigma\sqrt{n}})\right) \to -\frac{t^2}{2}.$$

Proof of CLT: old method

- In 1738, de Moivre proved that (scale) binomial distribution $\frac{B(n,p)}{\sqrt{np(1-p)}}$ can be approximated by a standard normal distribution.

$$\binom{n}{k} p^k (1-p)^{n-k} \simeq \frac{1}{\sqrt{2\pi np(1-p)}} e^{-\frac{(k-np)^2}{2np(1-p)}}.$$

Theorem

We have

- $\sqrt{n}\frac{\overline{X}-\mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1)$.
- $\sqrt{n}(\overline{X}-\mu) \xrightarrow{d} \mathcal{N}(0,\sigma^2)$.
- $\overline{X} \overset{d}{\simeq} \mathcal{N}(\mu,\sigma^2/n)$

1 Central Limit Theorems

2 Applications

Applications

- We have

$$\mathbb{P}[Z_n = \sqrt{n}\frac{\overline{X} - \mu}{\sigma} \in [a, b]] \simeq \Phi(b) - \Phi(a) = \int_a^b \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx.$$

or,

$$\mathbb{P}[A \leq \overline{X} \leq B] \simeq \Phi(\sqrt{n}\frac{B - \mu}{\sigma}) - \Phi(\sqrt{n}\frac{A - \mu}{\sigma})$$

Examples

- Suppose $X_n \sim Poisson(10)$ for $n = 1, \ldots, N$, and let $\overline{X}_N$ be the sample average. Use the Central Limit Theorem to approximate $\mathbb{P}[9 \leq \overline{X}_N \leq 11]$ for $N = 20$.
- We have $\mu = \mathbb{E}[X_n] = 10$ and $\sigma^2 = \mathrm{Var}(X_n) = 10$. By CLT, $\mathbb{P}[9 \leq \overline{X}_N \leq 11] \simeq \mathbb{P}[9 \leq \mathcal{N}(10, 10/20) \leq 11] = \Phi(\frac{1}{\sqrt{0.5}}) - \Phi(\frac{-1}{\sqrt{0.5}}) = 0.92140.0786 = 0.8427.$
- We can calculate exactly since $20\overline{X} \sim Poi(200)$ and

$$\mathbb{P}[9 \leq \overline{X}_N \leq 11] = \mathbb{P}[180 \leq 20\overline{X}_N \leq 220] = 0.8425.$$

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Examples

Suppose you have collected $N = 100$ data points from an unknown distribution. The only thing you know is that the true population mean is $\mu = 500$. and the standard deviation is $\sigma = 80$. (Note that this distribution is not necessarily a Gaussian.)

a). Find the probability that the sample mean will be inside the interval $(490, 510)$.

b). Find an interval such that 95% of the sample average is covered.

$$\overline{X}_N \overset{d}{\simeq} \mathcal{N}\left(500, \left(\frac{80}{\sqrt{100}}\right)^2\right).$$

$$\begin{aligned} a.\ \mathbb{P}[490 \leq \overline{X}_N \leq 510] =& \Phi\left(\frac{510-500}{80/\sqrt{100}}\right) - \Phi\left(\frac{510-500}{80/\sqrt{100}}\right) \\ =& \Phi(1.25) - \Phi(-1.25) = 0.7888 \end{aligned}$$

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b. Find an interval such that 95% of the sample average is covered. Hint: $\Phi(0.975) = 1.96$, i.e. 95% values of $\mathcal{N}(0,1)$ is in the interval $[-1.96, 1.96]$. Hence

$$\frac{y - 500}{80/\sqrt{100}} = \pm 1.96 \Rightarrow y = 484.32 \quad \text{or} \quad y = 515.68$$

Therefore

$$\mathbb{P}[484.32 \leq \overline{X} \leq 515.68] = 95\%.$$

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Confidence interval

- In applied statistics, normally, we need to estimate μ (unknown) with respect to some confidence $\alpha = 95\%$ (or $99\%, 90\%$).
- Idea: with proba 95%,

$$\overline{X} \in \left[\mu - 1.96\frac{\sigma}{\sqrt{n}}, \mu + 1.96\frac{\sigma}{\sqrt{n}}\right] \Rightarrow \left|\overline{X} - \mu\right| \leq 1.96\frac{\sigma}{\sqrt{n}}$$

Hence, if from the data, we get $\overline{x}$, then with proba 95%

$$\mu \left[\overline{x} - 1.96\frac{\sigma}{\sqrt{n}}, \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right].$$

- The interval $\left[\overline{x} - 1.96\frac{\sigma}{\sqrt{n}}, \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right]$ is the 95% confidence interval.
- The tolerance error of the estimation is $1.96\frac{\sigma}{\sqrt{n}}$. $\Rightarrow$ For given error ϵ, choose the suitable sample size N.

- Let $X \sim B(100, 1/2)$. Find the probability $\mathbb{P}(X = 50)$ and $\mathbb{P}(X \leq 56)$.
- Normally,

$$\mathbb{P}(X = 50) = \frac{\binom{100}{50}}{2^{100}}, \quad \mathbb{P}(X \leq 56) = \mathbb{P}(X = 0) + \ldots + \mathbb{P}(X = 56)$$

Tendious calculation!!!

- By CLT, $X = Z_1 + Z_2 + \ldots + Z_{100}$

$$X \overset{d}{\simeq} \mathcal{N}\left(np, np(1-p)\right) = \mathcal{N}\left(50, 25\right).$$

Then

$$\begin{aligned}\mathbb{P}(X = 50) \simeq & \mathbb{P}[49.5 \leq X \leq 50.5] \simeq \Phi\left(\frac{50.5 - 50}{5}\right) - \Phi\left(\frac{49.5 - 50}{5}\right) \\ = & \Phi(0.1) - \Phi(-0.1) = 0.5398 - 0.4602 = 0.0796.\end{aligned}$$

$$\mathbb{P}(X \leq 56) \simeq \Phi\left(\frac{56 - 50}{5}\right) = \Phi(1.2) = 0.8849.$$

Remark

- If the ground distribution is not normal, then we only have a limit, but not an equality. Therefore, the approximation is only good if the sample size is LARGE enough.
- In practical statistics, we demand $n \geq 30$. For binomial distribution, both np and $n(1-p)$ are at least 5.
- Berry-Esseen bound,

$$\sup_{z \in \mathbb{R}} |F_{Z_n}(z) - F_Z(z)| \leq \frac{C\mathbb{E}[|X_1|^3]}{\sigma^3 \sqrt{N}},$$

for $C = 0.4748$.
- **Weak convergence**: $Z_n \xrightarrow{w} Z$ if for any bounded and continuous function $f : \mathbb{R} \to \mathbb{R}$,

$$\lim_{n \to \infty} \mathbb{E}[f(Z_n)] = \mathbb{E}[f(Z)].$$

- Relation

$$a.s \text{ or } L^p \Rightarrow \quad \mathbb{P} \quad \Rightarrow \; d = w.$$

Homework

Chapter 6. 19,20,21
Chapter 9. 2,3,6,7

Thank you!